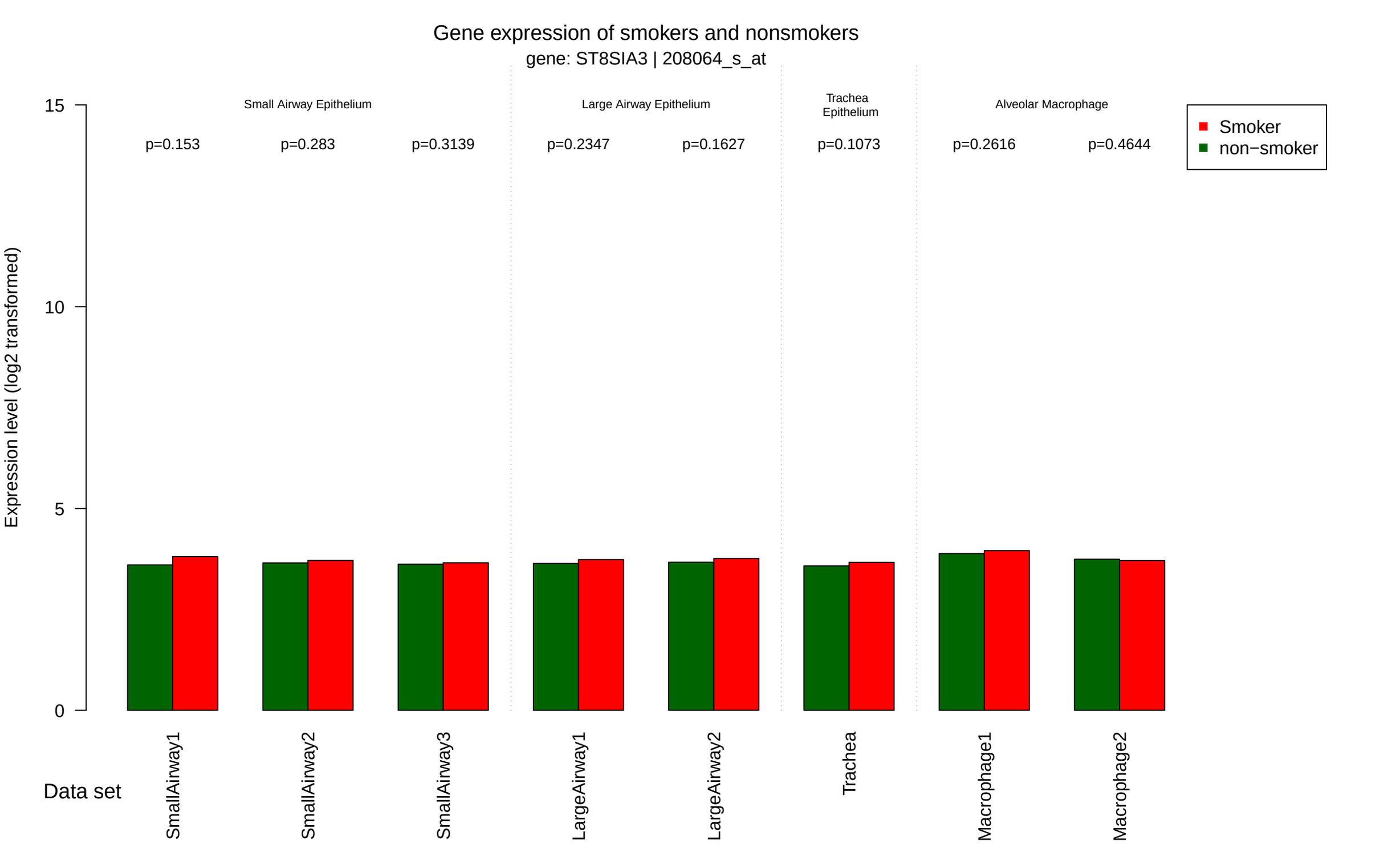
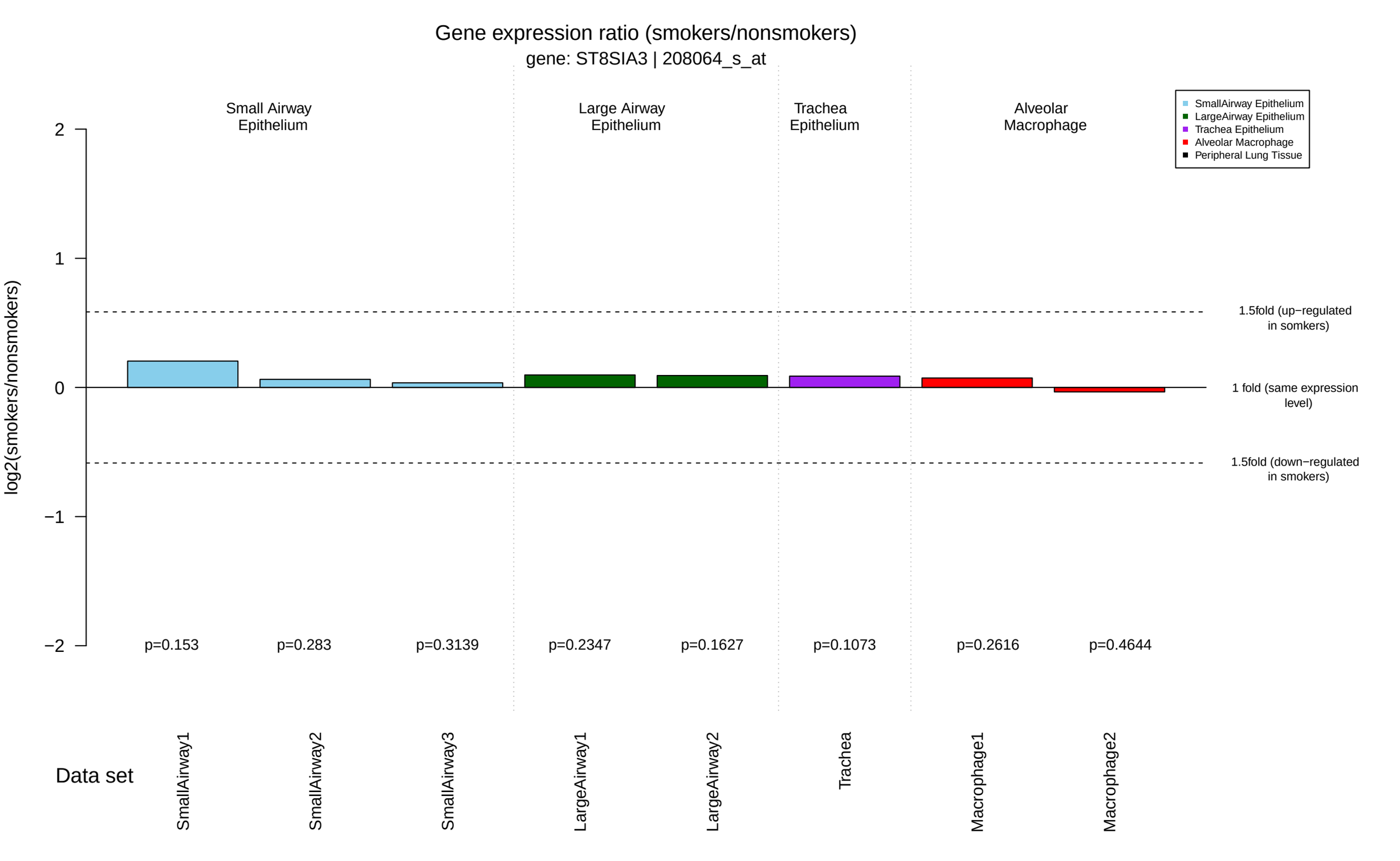
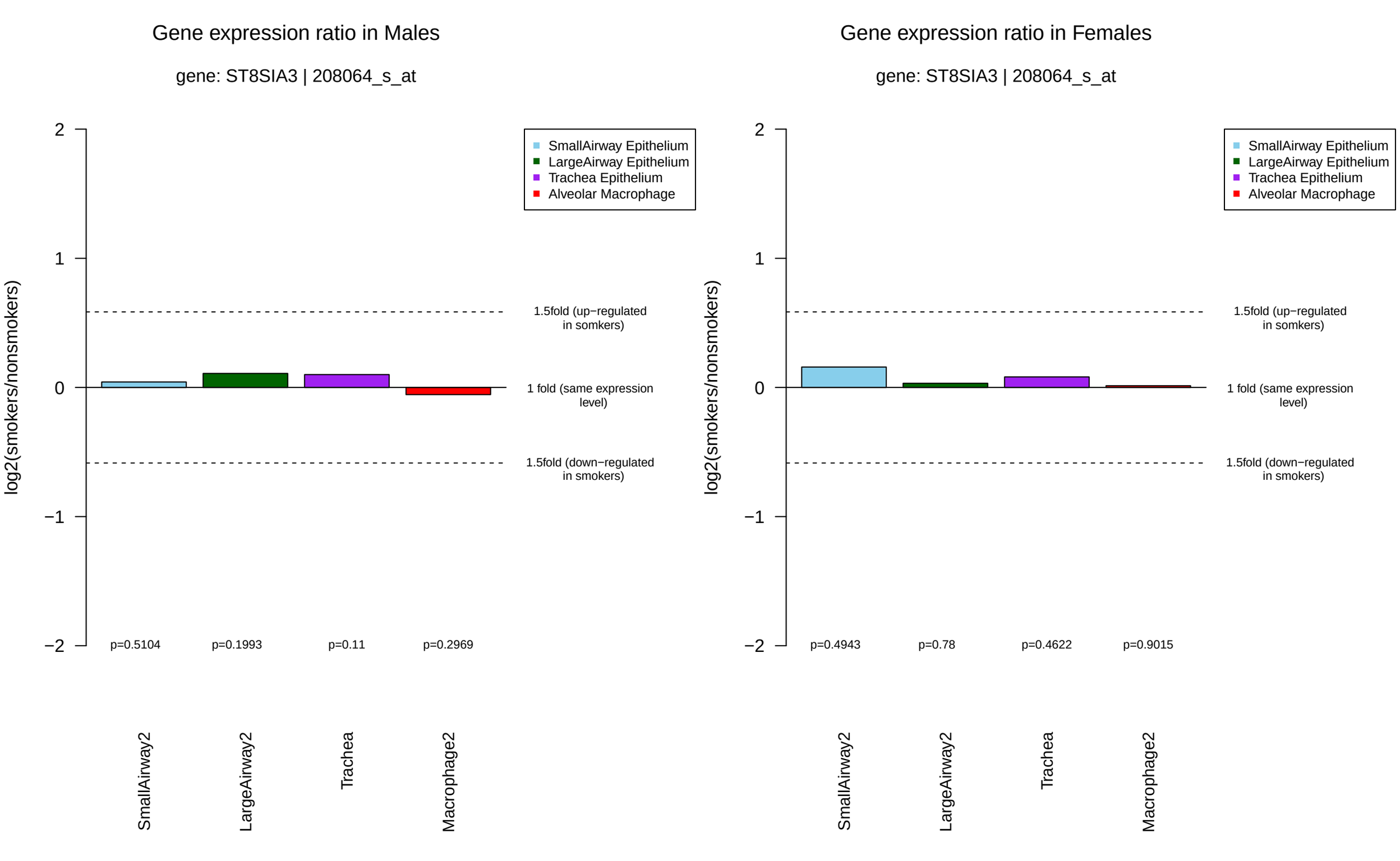
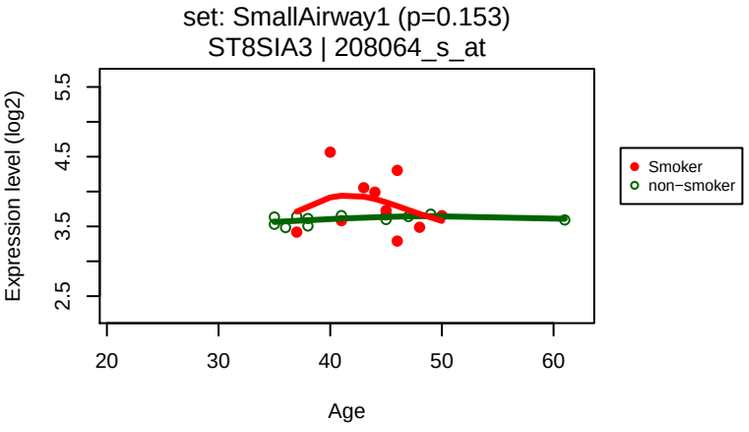
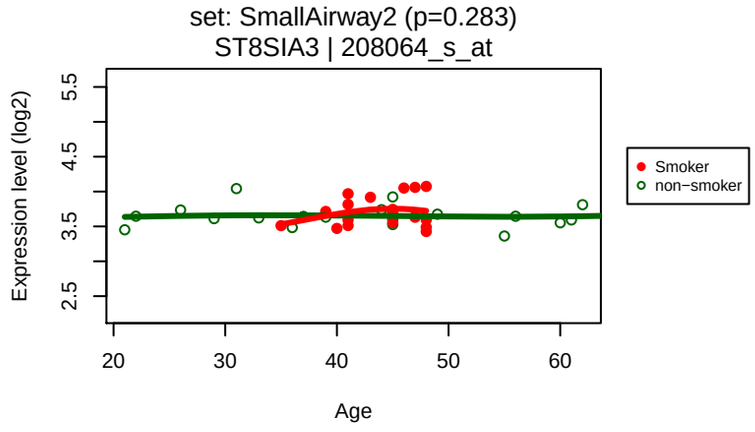
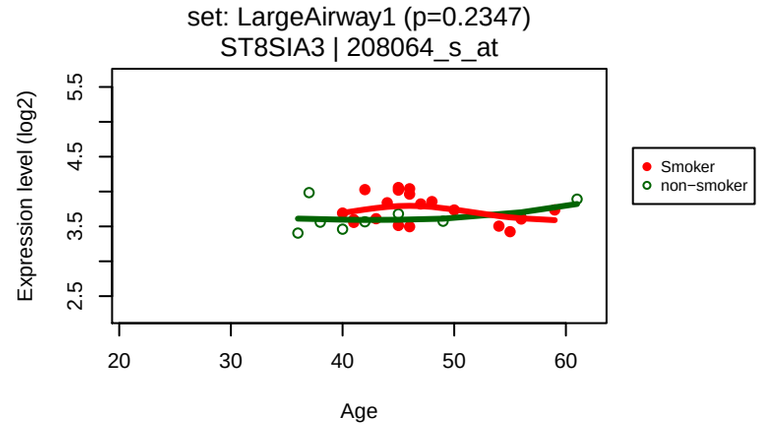

Gene expression of smokers and nonsmokers
gene: ST8SIA3 | 208064_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.153
p=0.283
p=0.3139
p=0.2347
p=0.1627
p=0.1073
p=0.2616
p=0.4644
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

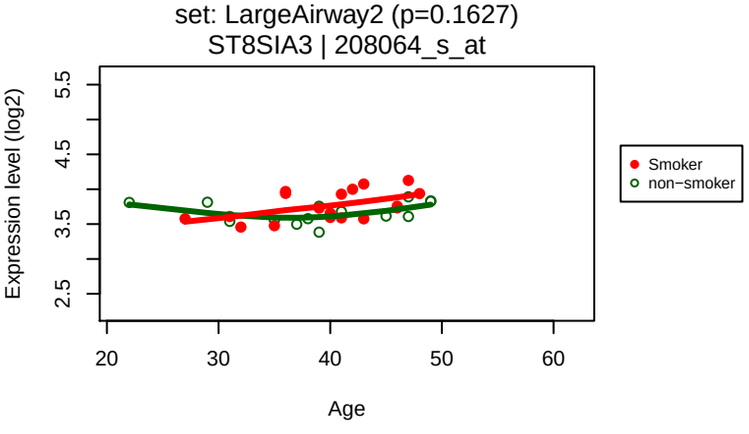

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA3 | 208064_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.153
p=0.283
p=0.3139
p=0.2347
p=0.1627
p=0.1073
p=0.2616
p=0.4644
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

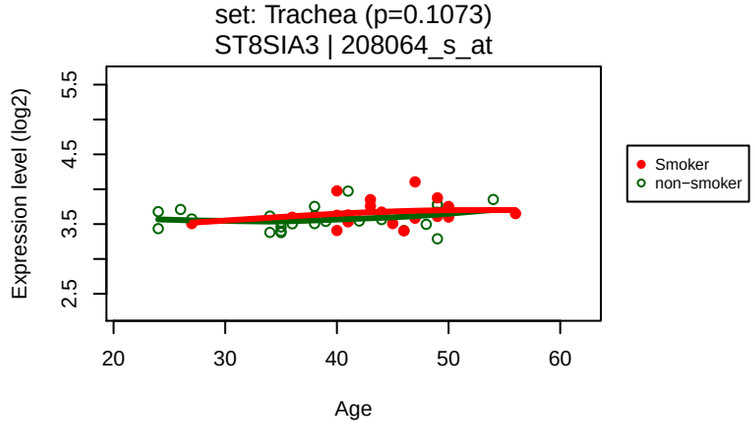

Gene expression ratio in Males
gene: ST8SIA3 | 208064_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5104
p=0.1993
p=0.11
p=0.2969
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA3 | 208064_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4943
p=0.78
p=0.4622
p=0.9015
SmallAirway2
LargeAirway2
Trachea
Macrophage2

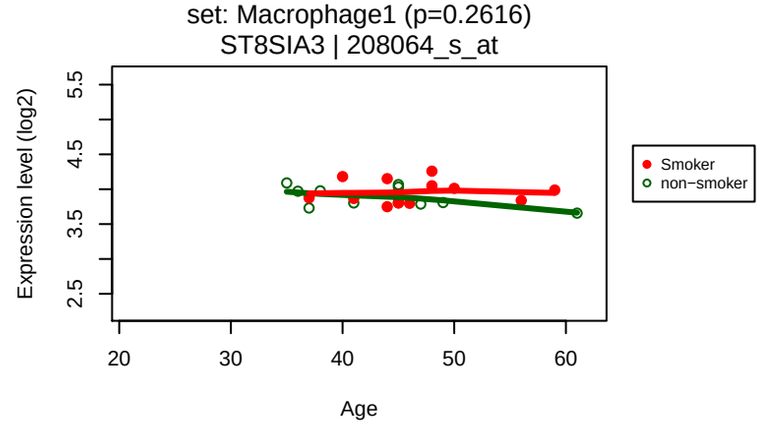

set: SmallAirway1 (p=0.153)
ST8SIA3 | 208064_s_at
Expression level (log2)
Age
Smoker
non−smoker

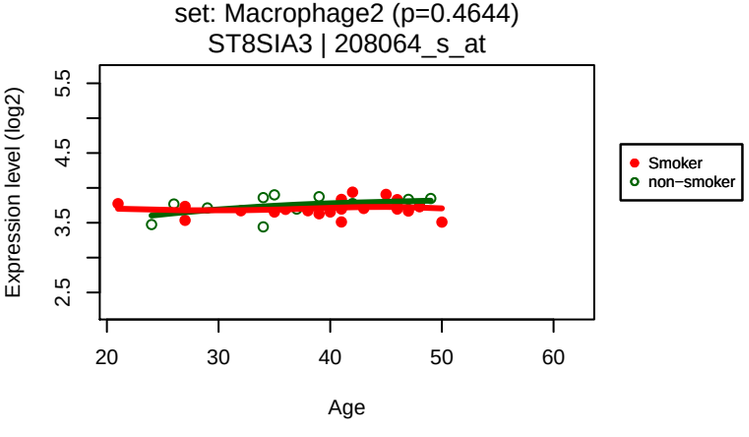

set: SmallAirway2 (p=0.283)
ST8SIA3 | 208064_s_at
Expression level (log2)
Age
Smoker
non−smoker

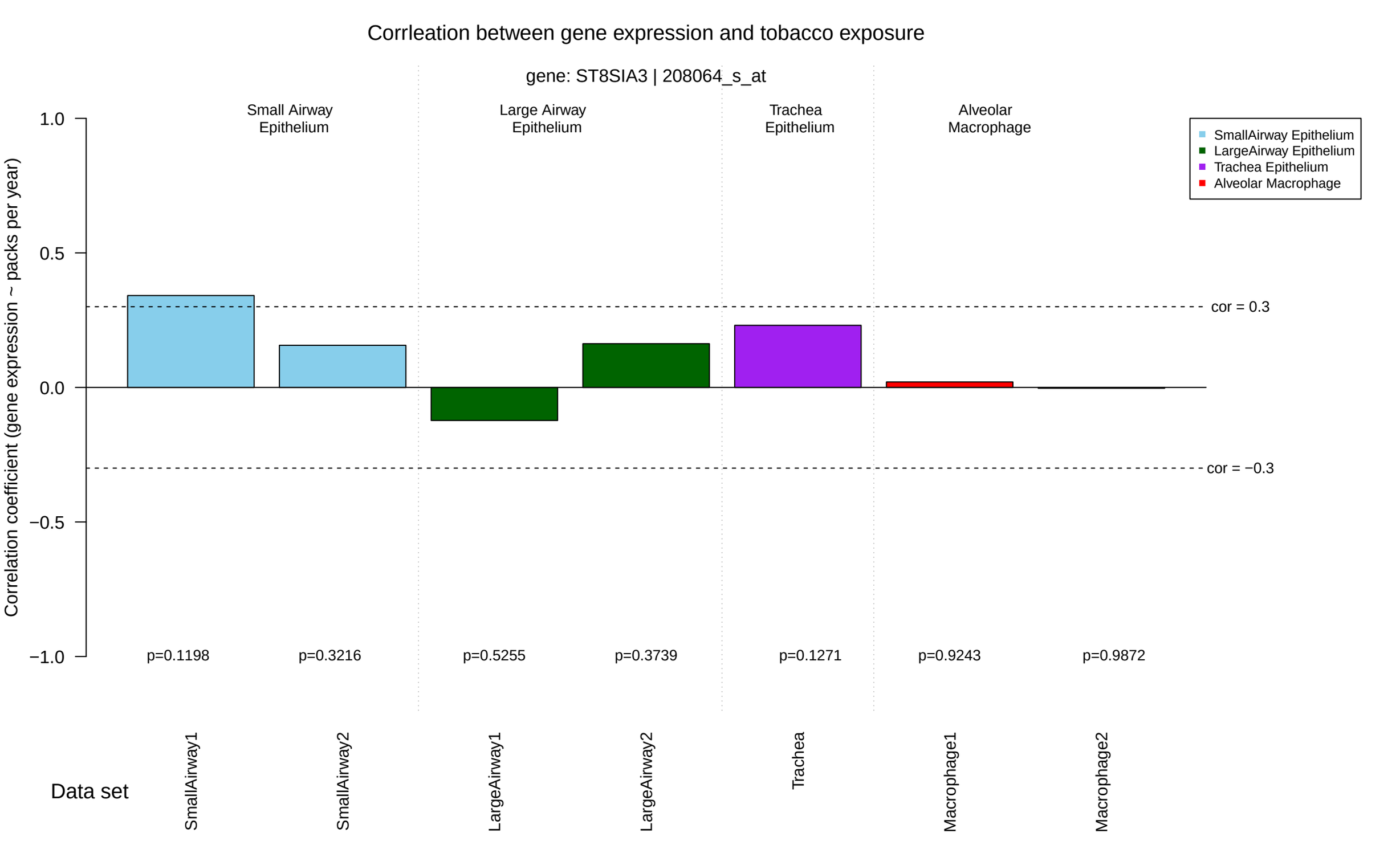

set: LargeAirway1 (p=0.2347)
ST8SIA3 | 208064_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1627)
ST8SIA3 | 208064_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1073)
ST8SIA3 | 208064_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2616)
ST8SIA3 | 208064_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4644)
ST8SIA3 | 208064_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA3 | 208064_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1198
p=0.3216
p=0.5255
p=0.3739
p=0.1271
p=0.9243
p=0.9872
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2